Concept2Cure.RI

TECHNICAL WHITE PAPER

# One governed chain across the product lifecycle

An AI-native, 21 CFR Part 11-governed regulatory platform spanning medical device and drug pathways — from grant funding and protocol design, through grounded authoring and validated agency transmission, to post-approval change control.

| | | | |
|---|---|---|---|
| **931** | **359** | **107** | **19,181** |
| TABLES FROM A BLANK DATABASE | REGULATORY AI TOOLS | PRODUCT SURFACES | TESTS PASSING |

**Concept2Cure, Inc.**
Prepared for prospective investors · August 2026
Assessed against the `concept2cure-v2` engineering branch

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Alongside its capabilities, this paper reports the severe findings of an independent readiness audit and the verified state of each one in the current code. Figures are derived directly from the repository; where a figure is drawn from that audit it is dated, because the audit was a snapshot and the platform has moved since. Forward-looking statements regarding roadmap, market conditions, and competitive position are estimates, not guarantees of outcome.

CONTENTS

# What this document covers

*The platform as it exists in code today — architecture, intelligence, compliance machinery, every capability domain, and the verified state of every severe finding an outside audit raised against it.*



A NOTE ON METHOD

Every quantitative claim is derived from the repository as it stands today. Figures drawn from the August audit are dated as such: that audit was a snapshot, substantial work has landed since, and its aggregate scores are reported as history rather than current state. A vendor selling into regulated life sciences is bought or rejected on whether its claims survive scrutiny, so these are written to be checked.

SECTION 01

# Executive summary

*Regulatory software is fragmented into four silos that no vendor spans. This platform is built as the spanning layer, across a lifecycle wider than the submission itself — and it has been tested against an independent audit, which is the most useful evidence available about how the team builds.*

### The market failure

Bringing a medical product to market means assembling evidence, drafting an argument from it, packaging that argument to an agency's exact technical specification, transmitting it through a validated channel, and then maintaining it for the commercial life of the product — with every step attributable and defensible years later. That work is split across four categories of vendor that do not talk to each other: evidence and literature platforms, AI authoring startups, publishing and validation incumbents, and regulatory information management systems. Sponsors buy three or four and stitch the seams with consultants.

### What has been built

A single governed system spanning that chain — and considerably more than the chain. Source documents enter a tenant-isolated evidence store; a retrieval-grounded assistant drafts against them with per-claim citations; the editor cannot save content lacking clause-level provenance; approval runs through an electronic-signature workflow; and the package is assembled to eCTD or eSTAR structure, validated, and handed to one of thirteen agency transports. Either side of that sit grant management, protocol development, ethics submissions, risk-based monitoring, clinical study reports, quality and QC, labeling, pharmacovigilance, and post-approval change control.

| 215 | 931 | 107 | 13 |
|---|---|---|---|
| SERVICE DOMAINS ~519K LINES | TABLES FROM A BLANK DATABASE | CLIENT SURFACES IN THE PRODUCT | AGENCY GATEWAYS |

### The recurring theme in the engineering

One doctrine shows up in module after module, and it is why the output is usable in a regulated setting: **the judgments a reviewer will challenge are computed, not generated.** The substantial-equivalence decision walks FDA's own flowchart deterministically. The SUPAC classifier is a pure function. Shelf-life is an ICH Q1E regression. Central statistical monitoring uses a robust modified z-score with no user thresholds. The protocol is *projected* from a structured design rather than authored, so the two cannot drift. Where the platform lacks data it refuses rather than approximating — and journey tests assert those refusals as passing outcomes.

### The audit, and what it produced

A 15-domain independent readiness audit, executed against a live instance on 10 August 2026, returned not ready for general availability. Its central observation was not absent features but *controls that reported success while doing nothing* — an integrity monitor verifying an empty chain as intact, RLS policies rendered inert by a superuser connection, an eCTD path emitting zero leaf files. In the four days that followed, **149 commits** landed as a direct remediation programme, and fourteen of the severe mechanisms it named were re-checked in the code for this paper and found closed. Section 24 gives the finding-by-finding state.

THE HONEST POSITION

The platform is pre-GA. Running production under enforced tenant isolation, consolidating the audit substrate, and raising request-path coverage are the live engineering items; a first filing additionally depends on licensed agency templates, validator licences, gateway credentials, and a qualification signed by a competent person. What is built is the part that is hard to build. What remains is substantially licences, credentials, signatures, and wiring — on a clock set by the standards migration in Section 02.

SECTION 02

# The problem and the market window

*The submission market is not underserved — it is mis-served, by vendors each solving one quarter of a problem that only makes sense as a whole.*

### The work, and where it breaks

A regulatory submission is an argument built from evidence, formatted to a specification, and defended under inspection. Four things must hold simultaneously: the evidence must be found and appraised; the argument must be written and cited; the package must conform byte-for-byte to an agency's technical standard; and every step must be attributable to a person and a moment. No single vendor delivers all four, so the handoffs between them are where sponsors lose months — re-keying content between systems, reconciling citations by hand, and discovering conformance failures only at the gateway.

### The timing: a mandatory migration reopens every stack

#### eSTAR v7.0 — mandatory since 3 Aug 2026

Every 510(k) and De Novo now flows through eSTAR. FDA technical-screening error rates fell from 20.5% to 5.4% under it. The *formatting* pain is shrinking; value is migrating to predicate strategy, testing rationale, and clinical evidence — the content problem.

#### eCTD 4.0 — the 2026–2029 window

PMDA mandatory April 2026, EMA around 2027, FDA around 2028–29. Every sponsor must migrate. Only about four vendors passed the EMA v4 pilot. This is the decade's one industry-wide drop in switching costs.

#### Gateway access is open

FDA's ESG NextGen retired the legacy WebTrader client in April 2025 in favour of a portal, REST API, and AS2. There is **no FDA certification programme for eCTD software** — validation criteria are published and a new entrant can offer direct submission on sponsor-held accounts after a short test phase.

#### AI is being normalized — and governed

FDA deployed its own generative system in 2025; FDA and EMA published joint AI guiding principles in January 2026; EU AI Act obligations phase in from August 2026. AI content is becoming acceptable, which turns *governance of the AI* into the procurement question.

### What that combination means

A buyer who must migrate to a new submission standard anyway, who is now permitted to use AI but must prove how it is controlled, and who currently pays three vendors to cover one workflow, is a buyer whose stack is genuinely re-openable. That condition does not persist. It closes when the migration completes and the incumbents' AI bolt-ons mature — which is the clock this business is running against, and the reason the remediation programme in Section 12 is sequenced for speed rather than elegance.

THE ONE THING THAT DOES NOT CHANGE

Whatever the standard or the model generation, a submission must be defensible: an inspector, years later, must be able to ask where a sentence came from and receive an answer. That requirement is indifferent to AI progress, and it is the requirement the platform's architecture is organized around. Frontier capability will keep improving for every competitor simultaneously; the ability to prove provenance will not arrive by upgrading a model.

Market facts are drawn from a competitive benchmark synthesized from public agency notices and industry research in August 2026, with per-claim sources retained in the platform's engineering documentation. Vendor capability descriptions reflect publicly stated positioning at that date; pricing is directional and is not reproduced here.

SECTION 03

# The competitive landscape

*Four silos, on both the device and the drug side, with the same structural boundary in each. The gap is not a niche — it is the middle of the workflow.*

| SILO | REPRESENTATIVE VENDORS | WHERE IT STOPS |
|---|---|---|
| **Evidence & literature** | Celegence CAPTIS, DistillerSR, CiteMed, Nested Knowledge, Basil Systems | Produces screened evidence and landscape data, not a submission-shaped argument. |
| **AI authoring** | Weave Bio, Peer AI, Yseop, Narrativa, Complizen, Cruxi, Essenvia | Stops at the document boundary — no assembly, validation, or transport. |
| **Publishing & validation** | Lorenz docuBridge, Extedo, Certara GlobalSubmit, Ennov | Validates and transmits; AI is a workflow bolt-on, not a drafting engine. |
| **RIM & lifecycle** | Veeva Vault RIM, Rimsys, RegDesk, Greenlight Guru | Tracks registrations, process, and quality; does not generate the content. |

### The notable positions

**Veeva Vault RIM — the gravity well**

Market leader with several hundred RIM organizations and a downmarket SKU. The durable advantage is distribution, not capability: recurring complaints concern speed at scale, a cumbersome data model, and services overhead, and its 2026 AI is workflow agents rather than drafting. This is the incumbent whose response determines the size of the window.

**Lorenz — the validation reference**

docuBridge plus eValidator, the industry's reference validation tool, and an FDA eCTD 4.0 pilot partner since 2022. Criticised on desktop-era usability. Relevant here as both a benchmark and a dependency: the platform carries an adapter seam for eValidator, and the licence is a procurement item.

**Weave Bio — the closest bridge**

AI IND drafting extended to NDA workflow in April 2026 via a contract-research channel, with a published sponsor case reporting a large time reduction and no critical errors. It has no assembly, validation, or gateway — the bridge is a services partnership, not a product.

**Greenlight Guru — the governance badge**

Medtech quality-system incumbent, ISO/IEC 42001 certified — first to make AI governance a marketable credential in this segment. It does not generate or fill eSTAR. Its certification is the clearest signal that *how the AI is governed* has become a purchasing criterion.

### The structural finding

Nobody spans intake → evidence → drafting → official package → post-market. The two closest moves are recent and partial: a RIM vendor added authoring-lite in May 2026 with no evidence engine behind it, and an AI authoring startup reached the drug workflow through a CRO channel. Meanwhile a publishing incumbent has stated publicly that it is evaluating the future of its regulatory writing unit — a signal that the services-attached model is under pressure from exactly this direction.

THE HONEST COUNTERWEIGHT

Every vendor above has something this platform does not: paying customers, executed submissions, and a validation history. Architecture spanning the chain is a real advantage only once it is trustworthy enough to be bought, which is what Sections 10 through 12 are about. The competitive claim in this paper is about product shape, not market position.

SECTION 04

# Platform architecture

*A single TypeScript monorepo organized in five layers, each of which the layer above can only reach through a governed interface.*

| | |
|---|---|
| **Surface** LAYER 5 | React 18 + Vite single-page application, **107 product surfaces**. **Chat-first**: capability is invoked by asking the assistant, not by navigating to a screen per feature. The canonical workspace is an asymmetric three-zone grid — contextual tree, intelligence stream, and a live-rendered artifact pane that never inherits chat state. A registry of 28 governed components is the single source of truth; raw form elements are rejected by a CI gate. |
| **Governance** LAYER 4 | The compliance spine (Section 09). Audit chaining, electronic signature with signing-authority resolution and version binding, document freeze and lock, clause-level lineage enforced in-transaction, and a default-deny authorization boundary mounted once ahead of all route registration — so no route file can forget authentication. |
| **Intelligence** LAYER 3 | AnA, the tool-calling assistant with **359 registered tools** (Section 06); the AI gateway every model call must pass through (Section 07); and the RIM, a deterministic non-LLM layer accumulating regulatory patterns (Section 08). |
| **Domain services** LAYER 2 | **215 service directories, ~519,000 lines** of regulatory subject matter: eCTD assembly and lifecycle operations, eSTAR generation, substantial-equivalence reasoning, clinical evaluation, study design, risk-based monitoring, biostatistics, CMC, quality management, labeling, pharmacovigilance, and thirteen agency transports. Exposed through **3,846 route handlers** in 465 modules. |
| **Substrate** LAYER 1 | PostgreSQL with pgvector, addressed through Drizzle — **931 tables, 787 row-level security policies** (Section 05). Object storage for source documents, Redis for cache and queueing, and a real-time collaboration backend for the editor. |

### Design decisions that matter commercially

**Fail-closed by default**

The server refuses to boot without its secrets and names what is missing. The AI gateway refuses to serve with no provider. The readiness endpoint fails closed on an incomplete schema. A regulated buyer's first question is what the system does when it is wrong.

**One of everything**

An enforced doctrine that there is never more than one implementation of a capability, backed by CI gates rejecting duplicate table definitions, unreachable migrations, and parallel runtime paths. It is why a system of this size stays modifiable.

**Refusal as a first-class result**

When the platform lacks a licensed template, a dataset, or a credential, it says so explicitly rather than approximating. Journey tests assert honest refusals as passing outcomes. A confident wrong answer is a liability here, not a feature.

WHY A MONOREPO OF THIS DEPTH IS THE ASSET

Regulatory subject matter does not compress. eCTD backbone construction, lifecycle operations, controlled vocabularies, region-specific validation profiles, substantial-equivalence logic, ICH conformance rules, and thirteen distinct agency transport protocols are each irreducibly detailed, and each must interoperate with the governance layer to be usable. The volume of domain code is not incidental scale — it is the barrier a point-solution competitor would have to cross to span the same chain, and it has already been paid for.

SECTION 05

# The data substrate and tenancy

*Multi-tenancy in regulated life sciences is not a feature — it is the precondition for being allowed to hold the data at all. This is the layer the August audit hit hardest, and the layer most changed since.*

### Provisioning

A single idempotent installer builds the schema from a blank PostgreSQL: **931 tables and 787 row-level security policies**, verified on a clean Postgres 16 with a distinct exit code on failure. It creates schemas and extensions, runs the table push, applies policies, records applied state in a ledger, takes a cluster-wide lock so two deploys cannot race, and re-verifies its own readiness contract — failing if the policy count is zero rather than reporting success on an unprotected database.

### Isolation

Policies use correct `USING` / `WITH CHECK` semantics with `FORCE ROW LEVEL SECURITY`, and an allowlist kept in sync by a CI gate. The audit's finding was that all of this was *inert*: the application connected as a table-owning superuser, which bypasses every policy. The remediation is a dedicated least-privilege `app_service` login role, provisioned by the installer with grants refreshed on every deploy, and a separate migration owner for DDL.

> **Stated plainly: provisioned, not proven**
>
> The role exists and the grants are applied. Running production under it — and demonstrating that a cross-tenant read returns zero rows with enforcement on — is the single most important remaining gate before a multi-tenant pilot, and the first item in the roadmap at Section 26. Until that is demonstrated, isolation should be treated as designed and provisioned rather than as operating.

### A CI gate for the class, not the instance

Two of the audit's isolation findings shared a root cause: data models with no tenant column. Rather than fix only the instances, a CI gate now rejects tenant-blind data models outright, so the next model to omit a tenant key fails the build rather than reaching production and being found by an auditor. This pattern — make the shape of the mistake fail loudly rather than fixing each occurrence — recurs throughout Section 24.

### Retrieval and vectors

Source documents are chunked and embedded into pgvector with project-scoped identity, so retrieval for a given project cannot reach another project's corpus, and canonical source identity means an upload resolves to one object rather than proliferating copies. This is what makes the citation guarantee in Section 06 meaningful: a claim traces to a chunk, the chunk to a document, and the document to the tenant and project that own it.

### Schema shape

Table families follow the lifecycle rather than the codebase's own structure — 27 tables for clinical study reports, 23 for protocol development, 14 for CMC, 12 each for risk-based monitoring and grants, plus families for IRB, supply chain, QMS, QC, device submissions, evidence, and the CTD backbone. Every table carries a primary key; the audit's judgment on the static schema was that it is an excellent skeleton, with the weakness in dynamic reality rather than design.

SECTION 06

# AnA — the regulatory assistant

*Not a chatbot beside a feature set. The invocation mechanism for the feature set.*

### The tool surface

**359 registered tools** across thirteen capability categories: submission and eCTD (32), evidence and intelligence (29), device and IVD (19), authoring and rendering (18), clinical (15), governance and data (11), quality management (10), CMC (9), nonclinical (9), biostatistics (8), clinical pharmacology (4), platform control (2), and the remainder covering adjacent regulatory operations. There is not one screen per tool — the UI renders the result and optionally navigates, which is why 359 tools sit behind 107 surfaces.

### Context assembly

Three memory layers are assembled before each turn: the working thread, project memory retrieved by vector similarity over stored knowledge atoms with confidence and importance weighting, and account-level context. Responses stream over server-sent events. Every call is recorded to an AI audit log with the resolved model, substrate, and region.

#### Grounding is checked deterministically

After each turn a self-verification round — no additional model call — extracts the verifiable identifiers from the answer (trial registrations NCT/ISRCTN/EudraCT, literature PMIDs and DOIs, FDA submission numbers across 510(k)/PMA/De Novo/NDA/BLA/ ANDA) and any quoted source text, and checks each against the evidence the assistant actually gathered that turn. Claims absent from that corpus are surfaced as unsupported. **The verdict is advisory — shown to the user, not used to block or rewrite.** Binding retrieval-verified citations into the document itself is Phase 2 work in Section 26.

### Intelligence questions and war-game auditors

Beyond single tool calls, the assistant carries **39 intelligence-question flows** — structured multi-step investigations for questions like CMC specification setting or IND submission readiness — and a set of **war-game auditors** that adversarially review a document as a specific reviewer archetype would: IND, NDA, BLA, CSR, CMC, 510(k) device, PMA, protocol, stability, labeling, safety-narrative, risk-management, briefing-book, and SOP auditors. The posture is deliberate: rather than asking a model whether a document is good, the platform asks a named reviewer persona what it would object to.

### Therapeutic-area profiles

**27 therapeutic-area profiles** supply the domain priors that make an answer specific rather than generic — the endpoints, comparators, and regulatory precedent that differ between oncology, cardiovascular, and rare disease. This is the difference between an assistant that can discuss a submission and one that knows what a reviewer in that area asks first.

#### WHY THE TOOL COUNT IS NOT THE MOAT

A competitor with the same model access can add tools. What is not quickly replicable is what the tools reach: 931 governed tables, deterministic domain engines whose outputs are citable, and an audit trail that makes their use defensible. The tool surface is the interface to the asset, not the asset.

SECTION 07

# The AI gateway and model governance

*One chokepoint every model call passes through, because in 2026 the procurement question is no longer whether AI may be used but how it is controlled.*

### The chokepoint

Every model call routes through a single gateway, and a CI gate rejects code that attempts to bypass it. The gateway performs provider routing and failover across multiple frontier providers, health tracking, cost accounting in cents against a per-organization budget, and a deterministic mode for testing.

### Two policy passes before dispatch

**Synchronous**

Token budget, role-aware prompt-injection scanning, blocked patterns, and rate limits. Injection scanning covers non-user content — retrieved documents and tool output — because an ingested PDF re-entering the loop as context is the realistic attack path, not a user typing an instruction.

**Asynchronous PII/PHI**

A governed content classifier over outbound content produces a structured verdict — allow, flag, redact, or block. For redaction it returns a copied message set safe for dispatch; the caller's messages are never mutated. Findings carry detector names, classes, and redacted excerpts, never the raw value, so they are safe to persist alongside prompt hashes.

### Placement — the enterprise unlock

Every provider is classified by three facts that together decide which compliance requirements a tenant can be served under: the **substrate** it runs on (shared frontier API, frontier-in-your-cloud via Bedrock, Vertex or Azure, or self-hosted), the **regions** it can be pinned to, and whether it carries a **zero-data-retention** guarantee.

| TENANT REQUIREMENT | WHAT THE GATEWAY WILL ROUTE TO |
|---|---|
| EU data residency | Only a provider whose placement includes the EU region, or self-hosted |
| Zero data retention | Only a substrate contractually capable of it — private cloud under a no-retention agreement, or self-hosted |
| Air-gapped | Only self-hosted inference |

The resolved substrate and region are recorded in the audit log for every call. Shared frontier providers are marked as *not* zero-retention by default, because enabling a real agreement is an operator action — a signed addendum plus a provider-side setting — rather than something the registry may assume.

**The provenance ledger, and how it failed**

The record of which model, at which temperature, from which prompt, on which substrate produced each output is the artifact an inspector asks for. The August audit found it recording nothing, for two independent and silent reasons: the table had no migration, and no call site passed the logger a database pool, so the write was guarded by a permanently false condition — while the gateway reported every call as audited. It now **fails closed at boot**: the platform refuses to start if the ledger is unreachable, on the reasoning that boot is the one moment where refusing to continue costs nothing already in flight. The per-call writer stays non-blocking, so an audit-store outage mid-flight cannot take inference down.

### What remains open

Model reproducibility is incomplete: dated model snapshots are not yet pinned across the board, and the exact request as sent is not fully recorded. Both are Phase 2 items in Section 26, and both are prerequisites for answering "reproduce this output" rather than "here is what we asked for".

SECTION 08

# The RIM — deterministic intelligence

*A layer that accumulates regulatory judgment over time, and is deliberately not a language model.*

### What it is

The Regulatory Intelligence Model sits beneath the assistant. It observes regulatory work as it happens and aggregates *signals* — recurring deficiencies, reviewer triggers, rejections, strong or weak language tendencies, data gaps, consistency issues, formatting problems, risk signals — into durable, tenant-scoped *patterns*. A pattern's occurrence count records how often it has been seen; its confidence rises with repetition but is bounded, so a single noisy loop can never assert certainty.

### Why determinism is the point

The store is pure: no network calls, no model, and no wall-clock dependency — identity and timestamps derive from inputs, so the same call sequence always yields the same state and tests get reproducible results. That is a prerequisite for anything influencing a governed document. Every run carries a unique identifier and is replayable; every signal carries provenance including framework version, pattern version, and run ID; persistence failures mark a run `degraded` rather than failing silently; and trend detection requires a minimum sample and version-compatible history rather than reading noise as a trend.

### What sits on top

**Readiness scoring**

A computed submission-readiness score with contributing factors exposed rather than a black-box percentage.

**Recommendation engine**

Next actions derived from observed patterns and the current state of the dossier.

**Evidence-confidence model**

An evidence chain with confidence attached, persisted through the data-lineage service.

**Change-impact propagation**

Cross-artifact and cross-module analysis of what a change breaks elsewhere in the dossier.

### Precedent and refuse-to-file patterns

Alongside the RIM sits a precedent-intelligence module (22 services) carrying approval precedent, refuse-to-file trigger patterns, and complete-response-letter trigger patterns. Where the RIM learns from a tenant's own work, this carries the industry's public record of what agencies have objected to — the two together being the difference between a system that checks a document against rules and one that checks it against what actually got rejected.

WHY THE RIM IS THE COMPOUNDING ASSET

Frontier model capability is available to every competitor on identical terms and improves for all of them simultaneously; no durable advantage rests on it. A deterministic, tenant-scoped, versioned record of what reviewers actually objected to — accumulated across customers and submissions — is not purchasable. Its value grows with usage rather than with the next model release. The caveat in Section 25 stands: the substrate exists, the outcome data that would make it compound does not yet.

SECTION 09

# Trust architecture — 21 CFR Part 11

*In regulated life sciences, output that cannot be defended under inspection has no value. This section states where each control genuinely stands, including the two the August audit found failing.*

| CONTROL | STATE | WHAT IS TRUE TODAY |
|---|---|---|
| **Electronic signature** | BUILT | Server-side credential re-verification, signing-authority and signer-role resolution, binding to a specific document version. The audit called the design right; consolidating to one system of record remains. |
| **Clause-level lineage** | BUILT | The editor cannot save content without clause-level provenance — enforced in-transaction and structurally gated in CI. A user can select any passage and ask where it came from. |
| **Audit chain** | IN FLIGHT | Chain enforcement exists on the reference store via an insert trigger that cannot be bypassed by an application path. The audit found the chain unpopulated and three integrity checks reporting an empty chain as intact. Checks now return an explicit `unverifiable` state; substrate consolidation and immutability grants are Phase 2. |
| **Tenant isolation** | IN FLIGHT | 787 policies, correct semantics, FORCE RLS — inert until recently behind a superuser connection. Least-privilege role provisioned; running production under it is the remaining step (Section 05). |
| **Authentication boundary** | BUILT | Default-deny gate mounted once ahead of all route registration. Algorithm-pinned JWT with CI-enforced rotation, bcrypt at 12 rounds, attempt-limited email OTP. The org-admin escalation the audit exploited is closed, as are the MFA token-class and SSO redirect bypasses. |
| **Qualification pack** | GENERATED | Eleven Part 11 controls mapped to the tests that exercise them, all eleven executing and passing, with a verdict that fails closed on any control lacking evidence. Execution and signature require a competent person. |

### The principle the audit tested

The audit's sharpest observation was not about any single control but that several failures shared a shape: *a control that reports success while doing nothing.* An integrity monitor verifying an all-null chain as healthy. A tenant-offboarding path reporting permanent deletion while the data survived. Governed AI commands returning past-tense success for work that never happened. In a regulated tool a silent-pass control is more dangerous than an absent one, because only the first gets relied on during an inspection.

WHAT CHANGED AS A RESULT

The remediation was not only to fix each defect but to make the class detectable: integrity checks that cannot verify now say so; the AI provenance ledger refuses boot rather than logging silently; a CI gate rejects tenant-blind data models. Fixing an instance closes one finding. Making the shape of the mistake fail loudly closes the ones not yet written.

A CI gate also scans the codebase for unqualified compliance claims. When an earlier build displayed "FDA 21 CFR Part 11 Compliant" on the public signup page, the platform's own gap register flagged it as unsupportable and it was softened to describe supporting Part 11 workflows. In a market where a compliance overstatement can invalidate a customer's filing, tooling that polices its own marketing language is commercially material.

SECTION 10

# What the platform spans

*The submission journeys are what customers buy first. They sit inside a span running from grant funding to post-approval change — and it is the stages either side of submission, where the lifecycle suites compete, that no submission vendor has touched.*

Every stage below is backed by tables and services in the codebase, not roadmap. The sections that follow take each in turn.

| | | | |
|---|---|---|---|
| **Funding**<br>Opportunities, proposals, awards, milestones, invoices, closeout | **Preclinical**<br>Nonclinical safety, first-in-human dose engine, M2.4 / M2.6 | **Study design §11**<br>23 protocol tables; design gates; protocol projected from structured design | **Ethics**<br>IRB submissions, sites, consent documents, reviews, reportable events |
| **Clinical ops §12**<br>Risk-based monitoring, site risk, central statistical monitoring, KRIs | **Supply**<br>Suppliers, materials, batches, shipments, cold-chain readings | **Study reporting §13**<br>27 CSR tables — the deepest family in the schema | **CMC §14**<br>Module 3 compiler, QbD, SUPAC, ICH conformance, shelf-life |
| **Quality §15**<br>QMS and QC — audits, training, OOS investigations, batch release | **Submission §16–18**<br>510(k) · CER/PER · IND/NDA — assembly and validation | **Transport §19**<br>13 agency gateways behind one governed transmit path | **Labeling §20**<br>SPL generation, prescribing information, SmPC QRD |
| **Post-approval §20**<br>CMC change classified against FDA and EMA rules | **Post-market §21**<br>E2B(R3) ICSR, vigilance, PMCF, registrations | **Correspondence**<br>Health-authority questions, agency meetings, CRL library | |

### Why the span is the argument

The same evidence recurs at every stage. An endpoint defined in a protocol reappears in the statistical analysis plan, the study report, Module 2.7, and eventually the label. In a fragmented stack each reappearance is a re-entry, and each re-entry can diverge — which is precisely what a reviewer looks for. A platform holding all the stages can make the later artifact a *projection* of the earlier one rather than a copy of it, so divergence becomes structurally impossible rather than a thing to be checked for.

Section 11 shows that pattern in its purest form: the protocol is not authored, it is rendered from the structured design, so the design and the protocol cannot disagree. Section 14 shows the same mechanism in CMC, where Module 3 sections carry the hash of the source they were compiled from and are marked stale when it changes.

A CAUTION ON READING THIS MAP

Breadth is not maturity. The stages differ substantially in depth: submission, CMC and study design are deeply built, while grants, supply chain and correspondence are thinner. The sections that follow state depth per domain, and Section 24 gives the independent view. Coverage of a lifecycle stage in this map means real tables and services exist for it — not that it is finished.

SECTION 11

# Study design — design as data

*The clearest expression of the platform's central idea: hold the design as a structured object, and render every downstream document from it rather than writing them.*

### The protocol is projected, not authored

A study design is held as a structured object — arms, endpoints, estimands, eligibility, schedule of activities. The ICH M11-structured protocol is then *projected* from that object, section by section, in harmonised protocol order. Because it is a projection it is honest by construction: each section renders only what the object actually contains, and where the object is silent the section is marked `partial` or `missing` with an explicit gap list — never filled with invented prose. Editing the design re-renders the protocol, so the two cannot drift.

The estimands and objectives reuse the same renderer the statistical analysis plan uses, so the protocol's objectives section and the SAP agree word for word. One source of truth, two documents, no reconciliation step.

### Design gates

Before a design can advance it must pass deterministic gates — pure functions over the design object covering estimands, endpoint architecture, design framework, sample-size and power red flags, and the schedule of activities. Nothing calls a database, a clock, a random number generator or a model, so the same design always yields the same findings. They are reproducible, unit-testable, and they never fabricate a value to make a design look better than it is.

| SEVERITY | CONTRACT |
|---|---|
| CRITICAL | Blocks design approval outright — for example a non-inferiority design with no margin |
| MAJOR | A reviewer-likely deficiency; blocks the QC-to-approved transition |
| MINOR | A defensibility weakness worth addressing |
| INFO | Advisory |

### Evidence-grounded priors

A trial simulator is only as honest as the prior it integrates over. The platform turns historical evidence — effect sizes read from prior-phase study reports, comparable registered trials, and the literature — into a prior distribution on the treatment effect using inverse-variance weighting with optional relevance down-weighting and a DerSimonian–Laird random-effects estimate of between-source heterogeneity.

Two design choices are worth naming. First, it **refuses to launder an assumption as data**: when no evidence is supplied the prior is marked `assumption`, deliberately widened, and every gap surfaced as a caveat. Second, disagreement between sources *widens* the prior rather than being averaged away, and what is returned is the predictive standard deviation — the uncertainty about the effect a *new* trial would see — not the smaller standard deviation of the pooled historical mean. That is the number the assurance calculation actually needs, and using the wrong one is the standard way a simulated trial flatters itself.

### What else is in the module

CRF shell generation, design validation, registration projection (the same design rendered as a trial-registry record), and a CSR evidence source that reads prior study reports back into design decisions — closing the loop between what a previous trial found and what the next one assumes.

SECTION 12

# Clinical operations and monitoring

*Risk-based monitoring under ICH E6(R3) — an entire product category in its own right, and one no submission vendor carries.*

### The engine

A deterministic scoring engine with RACT, KRI and QTL seed libraries, aligned to ICH E6(R3) risk-based quality management and ICH E8(R1) quality-by-design critical-to-quality factors, with default catalogs following the TransCelerate RACT model. The same pure functions back the API routes, the site-risk engine, and the assistant's tools, so a risk assessment run from chat and one run from the UI cannot disagree.

> **"Not evaluated" is a first-class state**
>
> A key risk indicator with no reading — or with no threshold to read against — has not been shown to be in control, so banding it green would assert an assurance the data does not support. The status enumeration therefore carries `not_evaluated` alongside green, amber and red, and quality tolerance limits carry it alongside within, approaching and breached. Callers must distinguish "measured and within range" from "never measured". This is a small design decision that tells you how the system thinks: an unmeasured control is not a passing control.

### Site risk

Per-site risk snapshots derive from site intelligence, with monitoring tier assigned risk-proportionately per ICH E6(R3) — higher risk drawing more oversight across reduced, standard and enhanced tiers. The engine is defensive by construction: two divergent site data shapes exist in the codebase, so it reads whichever score columns are present and treats a missing table as "no sites" rather than failing or inventing.

### Central statistical monitoring

A deterministic, unsupervised cross-site outlier engine. The premise from ICH E6(R3) central monitoring is that site-level data should be broadly comparable, so a site whose metric deviates sharply from the study cohort is a candidate risk. Each site is scored against the cohort distribution with a **robust modified z-score** — median and median absolute deviation, the Iglewicz–Hoaglin formulation — falling back to a classic mean-and-standard-deviation z-score when the MAD is degenerate.

Two properties matter commercially. There are **no distributional assumptions and no user thresholds** — the data picks the outliers, which removes the argument about where a threshold was set. And the functions are pure, with the route and assistant layers feeding in metrics and persisting signals, so a monitoring finding is reproducible from its inputs months later when a sponsor asks why a site was escalated.

### Where this sits competitively

Central statistical monitoring is a category with established specialist vendors. The platform does not claim parity with a dedicated CSM product. What it claims is that the monitoring signal, the protocol it monitors against, the study report it feeds, and the submission that report supports all live in one governed system — so a quality issue found in monitoring is traceable to the protocol section it violates and the submission section it will affect.

SECTION 13

# Study reports and the evidence spine

*Clinical study reports are the deepest data model in the platform — 27 tables — because the CSR is where a trial becomes an argument.*

### The CSR data model

Rather than treating a study report as a document with sections, the platform decomposes it into structured clinical content: studies, treatment arms, populations, eligibility criteria, endpoints and endpoint results, adverse events, safety summaries, pharmacokinetics, dose-response, biomarkers, statistical analyses, tables and figures, and references. On top of that sit cross-study comparisons, safety signals, and a knowledge graph of nodes and edges linking findings across studies.

The consequence is that a study report is queryable rather than merely readable. An endpoint result is a row, not a sentence — so it can be compared across studies, fed into the evidence prior for the next trial's design (Section 11), and projected into Module 2.7 without re-keying.

### The evidence spine

A separate clinical-regulatory-evidence module holds the org-scoped write and read path for evidence sources, clinical studies, regulatory findings, regulatory outcomes, relationships between them, and design lessons. Three rules are enforced by construction rather than convention:

- **Visibility boundary.** Reads see globally-public evidence plus the caller's own organization; writes are stamped with the caller's org unless an explicit governed global write is requested.
- **Inferred relationships** are marked as inferred rather than asserted as observed.
- **No single-source generalized lesson.** A design lesson cannot be generalized from one source — the module refuses to turn one trial's outcome into a rule.

That third rule is unusual and worth dwelling on. The commercial temptation in this category is to generalize aggressively, because confident-sounding lessons demonstrate value. Refusing to generalize from a single observation costs apparent capability and buys defensibility — which is the same trade the platform makes throughout.

### Complete response letters and refuse-to-file

The evidence spine ingests complete response letters, and the precedent module carries CRL trigger patterns and refuse-to-file trigger patterns. Combined with the war-game auditors in Section 06, this is the platform's answer to the question a sponsor actually asks: not "is this document well-formed" but "what will they come back on".

HONEST SCOPE

The CSR data model is deep and the extraction pipeline that populates it from source documents is the area the August audit found most defective at the ingest end — two content-hash defects and a broken entry point, both documented and scoped in Section 24. The model is strong; filling it reliably from arbitrary source documents is not yet finished.

SECTION 14

# Chemistry, manufacturing, and controls

*CMC is where a submission meets manufacturing reality. It is also where a one-time filing becomes a continuing relationship — every manufacturing change for the life of the product has to be classified, justified, and reported.*

### The Module 3 compiler — the provenance argument, concretely

Source objects — specifications, analytical methods, stability studies, batch records, manufacturing processes, container-closure and comparability data — compile deterministically into the CTD 3.2.S drug-substance and 3.2.P drug-product subsections. Every compiled section carries a content hash, a staleness flag with a stated reason, and **per-section lineage recording which source object produced it and that source's hash at the moment of compilation**.

That last property is the platform's central claim made specific: when a stability study or a specification changes, every Module 3 section derived from it is marked stale *and told why*, rather than silently drifting out of date until a reviewer finds the inconsistency. Elsewhere in this paper provenance is an architectural principle; here it is a compiler.

### Deterministic engines, not model calls

#### SUPAC / variations classifier

A pure function — no database, no model, no side effects. A proposed manufacturing change returns an FDA reporting category (Annual Report, CBE-0, CBE-30, or prior approval), a SUPAC tier, an EMA variation class (IA / IAIN / IB / II), bioequivalence requirements, impacted CTD sections, and citations to the controlling guidance. Grounded in SUPAC-IR/MR/SS, 21 CFR 314.70, ICH Q12, and EC No 1234/2008.

#### Shelf-life estimation — ICH Q1E

Ordinary least-squares fit of the stability attribute against time, solved to where the one-sided 95% confidence limit meets the specification limit, with the t-quantile evaluated against a tested Student-t distribution so the result reproduces exactly. The code states its own scope limit: single attribute, single factor; batch poolability is a separate module rather than an implied capability.

#### ICH compliance checker

Deterministic evaluation of a project's real stored data against Q1A(R2), Q2(R1), Q3A/Q3B, Q3D(R2), Q6A/Q6B, Q8(R2), Q9, and Q10 — the guidelines most frequently cited when FDA refuses to file.

#### Quality-by-design analyzer

Derives critical quality attributes from specification test parameters, impurity profiles, and stability indicators, and critical process parameters from manufacturing process records and in-process controls — from the project's actual data, replacing type-string heuristics wherever a real project exists.

WHY THIS MODULE CARRIES COMMERCIAL WEIGHT

A submission platform sells once per filing. A change-control system is used continuously for the commercial life of the product: every process change, site transfer, supplier substitution, and scale-up must be classified against both FDA and EMA rules and reported in the right category. The change-control store is org-scoped and real, and its verdicts are *computed by the classifier at read time rather than stored* — so a change in the rules re-projects every historical change rather than leaving stale conclusions in the database. That is the difference between a submission tool and a regulatory system of record.

SECTION 15

# Quality management and quality control

*The quality system is the thing a regulated customer already owns and will not replace lightly — which makes it both the hardest sale and the strongest retention surface.*

### What is built

**Quality management system**

Controlled documents, training records, supplier management, internal audits, management reviews, and nonconforming product — the ISO 13485 and 21 CFR 820 spine. A change-control service and SOP templates sit alongside.

**Quality control**

Specifications, out-of-specification investigations, batch release, deviations, microbiological testing, and reference standards — the laboratory and release side, which is where GMP inspections concentrate.

### Why OOS investigations matter disproportionately

An out-of-specification result is the single most inspected workflow in a GMP quality system, because it is where the incentive to conclude "laboratory error" is strongest and where FDA's expectations are most explicit. Holding OOS investigations in the same governed store as batch release, the specification they failed against, and the CMC change control that may have caused them, is a materially different proposition from holding them in a standalone eQMS that knows nothing about the submission.

### The connection to CMC and submission

A specification lives in one place and is referenced by the QC test that runs against it, the Module 3 section that describes it (Section 14), and the change control that modifies it. When the specification changes, the QbD analyzer re-derives the critical quality attributes, the Module 3 section is marked stale with its reason, and the SUPAC classifier determines what reporting category the change requires. In a fragmented stack those four systems are four separate updates, performed by different people, reconciled by hand.

### Honest depth

This is a thinner module than CMC or submission — three core services plus the schema — and it should be read as the quality spine that connects the deeper modules rather than as a replacement for an established eQMS. The competitive observation in Section 03 applies here with force: Greenlight Guru and Qualio own this category, they are entrenched, and the argument for displacing them is integration with the rest of the lifecycle rather than feature parity.

TRAINING RECORDS, AND WHY THEY APPEAR HERE

Training records look administrative until an inspection, at which point "was this person qualified to perform this action" is asked about every signature in the audit trail. Because the platform holds the training record and the electronic signature in the same governed store, that question is answerable by query rather than by retrieval from a separate system — which is the kind of small integration that decides an inspection day.

SECTION 16

# Device — 510(k), De Novo, PMA

*The 510(k) is decided by one intellectual step — is this device substantially equivalent to a predicate — and the platform implements that step as a deterministic, auditable walk rather than a generated opinion.*

### Substantial equivalence as a decision walk

Predicate tooling that finds candidates and stores a similarities-and-differences table is common. What is uncommon is making the SE *decision*. The platform implements FDA's own decision flowchart from *The 510(k) Program: Evaluating Substantial Equivalence in Premarket Notifications* as a deterministic walk returning `SE`, `NSE`, or `INSUFFICIENT_DATA` — with the exact decision path taken and the rationale at each node.

> **What it will not do**
>
> The boolean judgments the flowchart depends on — does the intended use match, do the differences raise new questions of safety or effectiveness — are **reviewer determinations the engine consumes, not values it fabricates**. The engine compares technological characteristics attribute by attribute to feed the "same technological characteristics" node, and then walks. Where a determination is absent it returns insufficient data rather than guessing. This is the difference between a tool that helps a regulatory professional reason and one that pretends to replace them.

### eSTAR

A versioned template catalog and registry pinned to the current eSTAR version with the retirement of the prior version recorded, a field map, content-leaf generation, a filing- readiness assessment, and a registration service. The design anticipates that template-chase is continuous rather than a one-time integration — FDA revises eSTAR, and a platform that treats the template as a fixed asset breaks on every revision.

Official template packs and field maps are licensed assets. The system **fails closed** without them rather than approximating a form it does not hold, which is the correct behaviour and also a procurement dependency (Section 26).

### Predicate and standards intelligence

Predicate adequacy assessment, predicate-intelligence tools available through the assistant, and a product-code to FDA-recognized-consensus-standards mapping. The standards mapping renders three distinct states — dataset not held, list holds nothing for this product code, or here is FDA's list — rather than collapsing absence and emptiness into a single silent result.

### Beyond 510(k)

Pathway engines cover PMA and device assembly manifests alongside eSTAR and PreSTAR, and a technical-file manifest engine serves the EU side. Device submission tables carry submissions, workflows, documents, profiles, and a device-specific audit trail. Design controls and human-factors surfaces address the design history file and usability engineering that a device submission must evidence.

### Position

Against the device-side field, the substantial-equivalence engine, versioned template registry, standards mapping and claim-to-source traceability are live, and multi-user authoring with Part 11 audit exceeds every device competitor assessed in the August benchmark. The gap is procurement — official templates and field maps — not engineering.

SECTION 17

# EU MDR and IVDR — CER and PER

*Notified bodies reject pure-AI clinical evaluations. That single market fact makes human-in-the-loop provenance the decisive capability rather than a compliance overhead.*

### The structure

Rule packs and a conformance validator implement the Annex XIV structure and MEDDEV 2.7/1 rev 4 expectations, with general safety and performance requirements handled by a dedicated GSPR module. Fourteen related table families cover evidence, literature, vigilance, sections, versions, compliance, workflows, exports, device profiles, and essential requirements.

### Literature and appraisal

A PubMed client feeds a screening and appraisal workflow whose decisions now **persist** — one current decision per organization, entry, programme and appraisal stage, with reviewer and timestamp, and a MEDDEV §8 exclusion rationale *enforced by a database constraint* rather than requested by a form. Stages use the CER flow's own title-abstract and full-text vocabulary rather than inventing new terms. Where an article is not in the corpus the endpoint refuses with an explicit status rather than fabricating a screening decision.

That constraint deserves emphasis. A literature exclusion without a documented rationale is the single most common notified-body finding in a clinical evaluation. Enforcing the rationale at the database level means the deficiency cannot be created, rather than being caught later by review.

### Vigilance and post-market

MAUDE and FAERS clients are real, with an honest posture on EUDAMED. The post-market module carries a PMCF plan generator, post-market authoring, readiness assessment, and PMCF enrolment — the periodic obligations that make a CER a recurring engagement rather than a one-time document.

### Export

Governed export renderers produce the notified-body-acceptable Word and PDF outputs, running through the same governed export path as every other regulated artifact, so an exported CER carries the same provenance and audit guarantees as a submission section.

WHY THIS JOURNEY MAY BE THE BEST COMMERCIAL ENTRY

CER work is recurring by regulation, the incumbent tools are services-coupled consultancies, and the market's stated objection to AI entrants — that a notified body will reject an AI-written evaluation — is answered directly by governed provenance rather than by argument. A sponsor can be shown, per claim, which screened article supports it and who approved the inclusion. The remaining work here is surfacing built services in the workbench rather than building engines.

### Honest state

Rule packs, conformance validator, governed export renderers, MAUDE/FAERS clients, and persisted screening decisions are real. Complaint and PMCF enrolment feeds have generators and documentation status but no backend behind them yet, and the evidence workbench surfaces are the active wiring work.

SECTION 18

# Drug — IND, NDA, and eCTD

*The largest single module after the assistant, and the one where the August audit found the most consequential gap — since closed.*

### Assembly

Fifty services cover backbone construction, lifecycle operations, study tagging, regional rules, controlled vocabularies, cross-reference resolution, checksum manifests, leaf source resolution and path safety, PDF/A detection and pipeline, bookmark generation, DTD and schema bundling, and a qualification harness pinned to published FDA validation criteria. Both eCTD 3.2.2 and eCTD 4.0 / RPS builders are carried — the latter mattering because of the 2026–2029 transition described in Section 02.

### The leaf-rendering gap, and its closure

The audit found the compile path the interface called declaring leaf rendering unimplemented and emitting zero files — a package that could never be transmitted. The platform now runs two honest paths:

| PATH | BEHAVIOUR |
|---|---|
| **Spine-backed** REAL PACKAGE | A programme resolving to a canonical submission with placed leaves compiles through the real generator: ICH v3.2.2 backbone, MD5 manifest, and genuine rendered PDF leaves. The response's XML backbone is the package's real index, and the leaf count is files actually materialized. |
| **Draft backbone** HONEST REFUSAL | A legacy-store project with no linked spine compiles a draft backbone over authored section text where every leaf carries `rendered="false"` and the response states exactly why no leaf file exists and where rendering actually runs. |

The renderer itself is deterministic: the same input yields byte-identical output, with fixed metadata and epoch dates and no object streams. That is not fastidiousness — it is what keeps the MD5 each granule's index records stable across re-renders, which is the eCTD checksum contract.

### IND lifecycle

Thirty-six services covering amendments and their persistence, annual reports, briefing books, cover letters with context, cross-references, action items, a cockpit view, and a checklist view assembler. This is the machinery that makes an IND a living file rather than a one-time submission — the drug-side equivalent of the post-approval change argument in Section 14.

### Validation

An FDA-criteria validation subset is built with an adapter seam ready for the industry-reference validator. Structural validation evidence is required by the transmit path (Section 19): missing evidence is treated as unknown, and unknown blocks. Licensing the reference validator is a procurement item.

SECTION 19

# Agency transport

*Transmission is the single irreversible action in the platform. Nothing can un-send bytes to an agency — so the ceremony that makes it safe lives in exactly one place.*

### Thirteen gateways, one path

FDA ESG, EMA CESP, PMDA, MHRA, Health Canada, TGA eBS, Swissmedic, NMPA, MFDS, ANVISA, CDSCO Sugam, and HSA PRISM. The FDA implementation is real protocol code against the current ESG technical specification: AS2 over HTTPS with mutual TLS, a PKCS#7 signed and encrypted MIME envelope with AS2-From/AS2-To headers and synchronous or asynchronous message disposition notifications, plus an SFTP path for very large submissions and organizations without AS2.

### The governed transmit ceremony

The history here is instructive. The ceremony originally lived inline in one HTTP handler, which meant any other caller wanting to transmit had to re-implement it — and the assistant's 510(k) command instead called a second service that could never reach the wire at all. So the ceremony now lives in one place and every caller passes through it:

1. The descriptor naming the bytes must be the **server-generated** one persisted by the tenant-scoped assemble route — not a client-supplied path.
2. Path syntax, namespace confinement, and durable-key confinement are checked.
3. **Internal structural validation evidence must be present and error-free.** Missing evidence is `UNKNOWN`, and unknown blocks.
4. A per-package active-transmittal lock prevents concurrent sends of the same package.
5. The gateway registry's human-authorization guard and pre-transmit checks must pass.

> **Why "unknown blocks" is the right default**
>
> The tempting behaviour is to treat missing validation evidence as "not yet validated, proceed with warning". For an irreversible action that reaches a regulator, absence of evidence must be treated as failure rather than as permission. This is the same doctrine as `not_evaluated` in monitoring (Section 12) and `unverifiable` in audit integrity (Section 09): the system distinguishes "checked and fine" from "not checked", and never renders the second as the first.

### Acknowledgement and integrity

Acknowledgement tracking, bundle-integrity verification, namespace management, a regional packager, and a validator registry sit alongside the transports. Assembled bundles and their acknowledgements are the record a sponsor needs when an agency disputes what was received.

### The market opening

There is **no FDA certification programme for eCTD software**. Validation criteria are published, and the gateway is open to registrants who pass a short test phase. A new entrant can therefore offer direct gateway submission on sponsor-held accounts without an incumbent's permission — which is why transport, usually a moat for the publishing vendors, is reachable here. What remains is operational: credentials, and the test-phase exercise itself.

SECTION 20

# Labeling and post-approval change

*Approval is not the end of the regulatory relationship. It is the beginning of the part that recurs — and the part a submission tool cannot serve.*

### Structured product labeling

An SPL generation service produces conformant FDA Structured Product Labeling XML from structured product attributes — name, NDC, ingredients, labeling text — keyed to the NLM standard LOINC section codes, and validates SPL documents for structural completeness. It is pure: no database, no network, no model, identical input to identical output. Alongside it sit a prescribing-information service and an SmPC QRD catalog for the European equivalent.

### Why labeling is the highest-stakes artifact

The label is the legally operative document. It is what the product may claim, what the prescriber reads, and what a plaintiff's expert reads years later. It is also derived from everything upstream — the endpoints in the study report, the safety summaries, the specifications. Holding labeling in the same governed store as those sources means a label claim can be traced to the study result that supports it, which is precisely the question asked in a labeling dispute.

### Post-approval change

Covered in depth in Section 14: CMC change control with verdicts computed at read time by the SUPAC and EMA variations classifier, so the reporting category for every historical change re-projects when the rules change. The commercial shape is worth restating here because it is the core of the recurring-revenue argument.

| MODEL | REVENUE SHAPE |
|---|---|
| Submission authoring tool | Sells once per filing. Value ends at acceptance. |
| Publishing and validation | Sells per sequence. Recurring but transactional and commoditizing. |
| **Lifecycle system of record** | **Used continuously for the commercial life of the product** — every process change, site transfer, supplier substitution, scale-up, label update, and periodic report. |

### Registrations and market maintenance

A global registration-intelligence module (48 services) with per-market specifications (25) and a registry bridge carries the multi-market registration picture — which products are registered where, under what conditions, and what maintenance each market requires. This is the surface that a RIM incumbent sells, and it is the surface where the platform's breadth argument is most directly competitive rather than complementary.

HONEST DEPTH

Labeling is five services and the SPL generator is a skeleton generator plus validator rather than a full labeling management system with negotiation history and version comparison against an approved reference. The registration module is broad but its depth per market varies. Both are real and both are shallower than CMC or eCTD.

SECTION 21

# Pharmacovigilance and post-market

*Safety reporting is the most time-bound obligation in the regulatory calendar — 7 and 15 day clocks that do not pause — and the one where a missed transmission is a regulatory event in itself.*

### E2B(R3) individual case safety reports

An IND safety report under 21 CFR 312.32 reaches FDA as an Individual Case Safety Report. One service classifies the 312.32 obligation; a separate composer turns the same intake adverse-event record into structured ICH E2B(R3) data elements and a deterministic escaped XML projection keyed by the official element identifiers — case administration, primary source, patient, reaction, drug, and narrative.

> **The honesty note in the code itself**
>
> The composer's own documentation states that it emits the E2B(R3) *data-element projection* — elements carrying their canonical identifiers so the case is gateway-ready and QC-able — and **not** the full HL7 V3 ICSR transport envelope with batch wrapper and OIDs, which the dispatch gateway adds. A completeness report surfaces any missing mandatory element before transmit. This is the kind of scoping statement that survives diligence: it tells a reviewer exactly what the module does and does not do, in the module.

### Narrative generation

The case narrative is composed by a shared deterministic safety-narrative writer that **never invents clinical detail**. A safety narrative is a place where a fluent language model would confabulate plausibly and dangerously — filling a gap in a case with a clinically reasonable detail that did not occur. Making the narrative writer deterministic and shared means the same case always produces the same narrative, and every element in it traces to a field in the case record.

### Transport and persistence

An ICSR gateway transport and transmission persistence layer sit alongside, so a case's submission state is recorded rather than inferred. Disproportionality analysis for signal detection is implemented in the pharmacovigilance compliance service, and the August audit assessed it as properly implemented.

### Device post-market

On the device side, the GSPR post-market module carries post-market surveillance services, PMCF plan generation and enrolment, post-market authoring and readiness. MAUDE and FAERS clients feed both sides. Periodic safety update reporting connects back to the CER obligations in Section 17.

> HONEST DEPTH
>
> Pharmacovigilance is a thin module by service count relative to its regulatory weight — the E2B composition and transport are real and carefully scoped, but this is not a full safety database competing with the established PV platforms. Its value here is integration: a safety signal connects to the study report that observed it, the label that describes it, and the CER that must account for it.

SECTION 22

# Scale and the module portfolio

*Scale alone is not an asset. Scale that stays modifiable, under mechanical invariant enforcement, in a domain that cannot be compressed — that is.*

### What exists, counted

| DIMENSION | MEASURE | NOTE |
|---|---|---|
| Service directories / files | 215 / 1,527 | Plus 264 loose top-level service modules |
| Lines in service layer | ~519,000 | Regulatory subject matter, not framework code |
| TypeScript modules / lines | 5,053 / 1.44M | ~1.64M lines across 7,805 files counting all languages |
| Tables provisioned from blank | 931 | Single idempotent installer, verified exit 0 |
| Row-level security policies | 787 | Correct USING / WITH CHECK semantics, FORCE RLS |
| Route modules / handlers | 465 / 3,846 | Behind a global default-deny boundary |
| Client surfaces | 107 | The product's actual screens |
| Registered AnA tools | 359 | Across 13 capability categories |
| Test files / assertions passing | 1,694 / 19,181 | Unit, route, database, security, golden-journey |
| CI guard scripts / workflows | 79 / 19 | Architectural invariants — see Section 23 |

### The largest domains

| DOMAIN | SVCS | SCOPE |
|---|---|---|
| **AnA assistant** | 229 | 359 tools, 39 intelligence-question flows, 27 therapeutic-area profiles, war-game auditors |
| **Regulatory core** | 77 | Canonical document store, lifecycle bindings, filing taxonomy, substantial equivalence |
| **Assistant orchestration** | 58 | Command execution, kernel, context routing |
| **eCTD** | 50 | Section 18 |
| **Global RI** | 48 | Multi-market registration intelligence |
| **Report engine** | 38 | Immutable sealed records, lineage-trace, portfolio and regional renderers |
| **Intelligence / RIM** | 37 | Section 08 |
| **IND lifecycle** | 36 | Section 18 |
| **Submission gateways** | 28 | Section 19 |
| **IVD knowledge** | 28 | Standards, legal, regulatory and scientific knowledge for diagnostics |
| **Market specifications** | 25 | Per-market submission requirements |
| **Statistics** | 24 | Group-sequential, BOIN dose finding, enrolment forecast, external control, assurance |

WHERE THE AUDITOR JUDGED THE DOMAIN WORK STRONG

Faithful regulatory outlines — PMA against 21 CFR 814.20(b), EU CTA against CTR 536/2014, MDR Annex II, CTD Modules 2–5 — judged better than most competitors' first attempts. Correct biostatistics: the Lan-DeMets solver reproduces textbook O'Brien-Fleming and Pocock boundaries exactly, and disproportionality analysis is properly implemented. Exemplary licensed-terminology handling, failing closed with an explicit licence requirement rather than risking copyright exposure.

SECTION 23

# Invariants enforced by machine

*79 CI guard scripts that do not check style. They check the architectural properties that would otherwise decay silently — which is what makes a system this size qualifiable rather than re-auditable on every release.*

### What the gates enforce

#### Governance integrity

No code bypasses the AI gateway. The editor's save path cannot skip the lineage gate. Governed export routes keep their consequence shape. Document aliases stay attribute-free. The completion ledger cannot claim done without evidence, cite a file that does not exist, or duplicate an identifier.

#### Tenancy and schema

Data models cannot be tenant-blind. The RLS allowlist stays synchronized across its consumers. No duplicate table definitions. Migrations must be reachable from a durable applier, and migration prefixes must not collide. Schema is diffed against a live baseline rather than against the repository itself.

#### Security posture

JWT verification stays algorithm-pinned. No development authentication path survives into production. No mock data reaches a production route. No committed secrets. Request values cannot reach a filesystem path unchecked. Password hygiene holds.

#### Product and interface

Compliance claims in the interface stay qualified. Design-system compliance and chip tones hold. Microcopy conventions hold. Client API calls must resolve to real endpoints. Design tokens meet contrast requirements. Stylesheets cannot orphan.

### Why this is unusual, and why it is worth money

Most CI enforces formatting and tests. These gates encode *architectural decisions* as executable rules, so a decision made once stays made. The gate that rejects tenant-blind data models is the clearest example: rather than fixing the two isolation defects an auditor found, it makes the entire class of defect unbuildable.

In a regulated product this is the difference between a system that can be qualified and one that must be re-audited on every release. A validation pack rests on the claim that controls remain in place between releases; mechanically enforced invariants are evidence for that claim in a form an auditor can execute.

#### WHERE THE QUALITY SYSTEM WAS ITSELF WEAK

The August audit's most consequential finding was about this layer. A global database mock installed across all test files meant the two database-backed, safety-critical CI jobs ran against a stub; measured request-path coverage sat at 5–13%; and no CI job ever ran in the RLS-enforced mode production mandates. **That is why the other blockers shipped green.** The mock is gone and golden-journey tests now drive all three regulatory journeys end to end, asserting honest refusals as passing outcomes — but raising coverage and making the browser tier real remain open, and the audit rightly called this the highest-leverage investment available, because it protects every other fix.

SECTION 24

# Independent audit and verified remediation

*The platform was audited across fifteen domains on 10 August 2026 by running it, and returned not ready for general availability. Every severe mechanism it named has since been re-checked directly in the code.*

| 14 | 149 | 19,181 | 132 |
|---|---|---|---|
| SEVERE MECHANISMS RE-VERIFIED CLOSED | REMEDIATION COMMITS IN FOUR DAYS | TESTS PASSING ACROSS 1,694 FILES | CI ACTIONS PINNED TO A COMMIT SHA |

### What the audit was

Each domain was audited by an independent pass that read the code, ran commands, and queried a live database, with every severe finding routed to an adversarial cross-check whose default was to refute. The worst were reproduced by hand against a from-scratch 931-table instance with the production build booted from `dist/`. Its central observation: several controls *reported success while doing nothing.*

### Finding by finding, as the code stands today

| WHAT THE AUDIT FOUND | NOW | VERIFIED IN THE CURRENT CODE |
|---|---|---|
| eCTD compile emitted zero leaf files | CLOSED | A deterministic renderer produces genuine, byte-stable PDF leaves; the spine-backed path assembles a real ICH v3.2.2 package (Section 18). |
| Production could not boot under enforced isolation | CLOSED | The startup posture probe runs inside a declared system tenant scope — it no longer trips the guard it should pass. |
| App connected as a superuser, making RLS inert | PROVISIONED | Least-privilege `app_service` role provisioned by the installer, grants refreshed each deploy. Running production under it is the remaining step (Section 05). |
| Any signup could delete any tenant | CLOSED | The org-admin escape hatch is scoped so a platform-role guard cannot receive a stand-in; tenant management sits behind platform-admin middleware. |
| MFA bypassable by password alone; SSO redirect takeover | CLOSED | Pre-MFA and refresh token classes rejected at every privileged route; SSO return targets and org selection validated. |
| AI provenance ledger recorded nothing | CLOSED | A boot invariant fails the platform closed if the ledger is unreachable, while the per-call writer stays non-blocking (Section 07). |
| Integrity checks called an empty chain healthy | CLOSED | An explicit `unverifiable` state with a stated reason; the Part 11 surface no longer asserts integrity it never checked. |
| Unencrypted database dump in the tree; DB TLS unverified | CLOSED | Dump removed and prohibited in the deploy script; TLS verification enabled and held by a regression test. |
| Global database mock across all tests | CLOSED | Gone from the shared setup; golden-journey tests drive all three journeys end to end (Section 23). |
| Fabricated programme data; frontend cancelled every timer | CLOSED | Invented programme identity removed from client data; the brute-force timer optimizer removed and client error reporting wired. |
| Supply chain and accessibility ungoverned | CLOSED | All 132 CI action references SHA-pinned, the AGPL dependency off the ingestion path, and committed-secrets and token-contrast gates in CI. |

WHAT THIS TABLE DOES AND DOES NOT ESTABLISH

Each row was confirmed by reading the current code. What has *not* been done is an independent re-scoring: the audit's headline counts were a 10 August snapshot, superseded by the work above, and no second audit has re-measured them. Treat the aggregate as history and this table as evidence. What remains genuinely open is in Section 26.

SECTION 25

# Defensibility and moat

*Five candidate advantages, sorted honestly by whether they compound, whether they are merely current, and whether they exist yet.*

**1 · Lifecycle span** BUILT

Grant funding through post-approval change, governed end to end, across device and drug. The volume of irreducible domain code is the barrier, and it is already paid for. *Durable but not compounding*: a well-funded competitor could cross it, but not quickly, and not without doing it twice.

**2 · Governed provenance** BUILT

Clause-level lineage enforced in-transaction; a save path that structurally cannot skip it; Module 3 sections carrying the hash of the source they compiled from. Retrofitting this into an existing authoring product means rewriting its persistence layer — which is why the AI authoring entrants are unlikely to add it and the publishing incumbents are unlikely to add drafting.

**3 · Deterministic domain engines** BUILT

SE decision walks, SUPAC classification, ICH Q1E regression, robust central statistical monitoring, design gates, evidence priors. Each is citable and reproducible — the property that makes output defensible under inspection, and the one a model-only competitor cannot assert.

**4 · Placement-aware inference** BUILT

Residency, retention, and air-gap constraints resolved per tenant at routing time. The prerequisite for large-pharma procurement. Currently differentiating, but *replicable* — expect it to become table stakes within two years as EU AI Act phase-in makes it a standard question.

**5 · Outcome data** NOT BUILT

Pairs of submitted content and the agency's actual response — what was filed, and what came back. The only advantage here that genuinely compounds, and the only one no competitor can purchase, because it accrues only to whoever holds the submission and the correspondence in one place. **Nothing captures it today.** The substrate that would make it trustworthy — the RIM, the evidence spine, the audit chain — exists; the capture does not. It is the highest-leverage open item on the roadmap.

### What is not a moat

Three things are worth naming as *not* defensible, because an investor will test them. The tool count is a capability inventory, not a barrier — a competitor with the same model access can add tools. The line count measures surface area, and the August audit demonstrated that surface area without coverage is a liability as much as an asset. And the design system, while genuinely good, is a matter of taste any well-resourced team can match. The defensibility argument rests on items 1 through 4 above, on the switching-cost window in Section 02, and eventually on item 5.

THE HONEST ASYMMETRY

Items 1 through 4 are built and real. Item 5 — the only one that compounds — is not. An investment here is a bet that the team converts a strong architectural position into the operational trustworthiness required to acquire customers, and then converts those customers into the outcome dataset that makes the position permanent. Both conversions are execution, and Section 24 is the best available evidence of how this team executes when told it is wrong.

SECTION 26

# Roadmap to general availability

*Sequenced so each phase unblocks the next and produces evidence a buyer's reviewers accept. A dependency order, not a wish-list — and the audit's own programme.*

| PHASE | GOAL | PRINCIPAL WORK AND ITS EVIDENCE GATE |
|---|---|---|
| 0<br>LARGELY DONE | **Stop the bleeding, make CI tell the truth**<br>Weeks 1–2 | Close the tenant-deletion path, rotate the exposed credential, remove the unencrypted dump, scope the test database mock, remove the timer optimizer, fix the SSO redirect and MFA token-class bypass. Gate: *every later fix becomes provable.* Verified closed in Section 24. |
| 1<br>IN FLIGHT | **Make isolation real, and make it boot**<br>Weeks 3–8 | Run under the least-privilege role; fix tables that would then deny-all; add residual tenant keys to route-reachable orphan tables; supply full environment in every deploy target; move submission output to durable storage; add Redis and leader election. Gate: *a CI job boots the built image in production mode and asserts readiness returns 200.* |
| 2<br>BEGUN | **Make the regulated core trustworthy**<br>Weeks 6–16 | Land chain and immutability on the durable applier; enforce immutability with triggers and revoked grants; serialize chain writes; consolidate to one signature system of record; replace advisory grounding with retrieval-verified citation binding; pin dated model snapshots and record the exact request. Gate: *an independent Part 11 gap assessment against the deployed schema.* |
| 3<br>NEXT | **Make one journey actually complete**<br>Weeks 10–22 | One eCTD pathway emitting a conformant sequence that passes an external validator; ship the review loop; correct filing-type to pathway mapping; complete the CER evidence workbench; repair the ingest-end extraction path; expose what is built in navigation; build the privacy lifecycle. Gate: *an automated golden journey that reopens the package and validates it.* |
| 4<br>BEFORE GA | **Earn the outside signatures** | Third-party penetration test; independent Part 11 gap assessment; restore-from-backup drill with documented objectives; raised coverage floors and a real browser tier; a VPAT passing WCAG 2.2 AA; retire or finish the route-shell modules rather than listing them as shipped. *Gate: evidence a buyer's security, quality, and regulatory teams accept on paper.* |

### What remains genuinely open

Audit-substrate consolidation and database-enforced immutability. Request-path test coverage and a real browser tier. Retrieval-verified citation binding. The privacy lifecycle — erasure, retention, and data residency. Outcome-data capture. Template-change ingestion. And proving production boots and serves under enforced isolation, which gates everything else.

### Procurement, running in parallel

Independent of engineering: official eSTAR templates and field maps, eCTD document type definitions, the reference validator licence, agency gateway credentials, and a medical terminology dictionary licence. These are single-source and the platform correctly refuses to approximate them — which converts part of the timeline from an engineering problem into a commercial one and should be underwritten as such.

REALISTIC TIMELINE

A friendly, single-tenant, caveated design-partner pilot is reachable at the end of Phase 1. A defensible first paying multi-tenant pharma customer — one whose quality and security teams sign off — is the full programme, estimated at 4–7 months with a 6–9 person team. The variance sits in validation and third-party evidence, which is calendar-bound rather than effort-bound: more engineers do not shorten a penetration test or a qualification signature.

SECTION 27

# Risks, and what to underwrite

*What could go wrong, stated at the same standard as the rest of this document.*

**Execution risk on trustworthiness**

The severe mechanisms an outside audit named are closed, but the regulated core is not yet through Phase 2 and no second audit has re-scored the platform. The risk is not that the work is unknown — it is scoped — but that hardening programmes routinely take longer than estimated, and this one runs on a clock set by the standards migration.

**Procurement dependency**

Licensed templates, validator licences, and gateway credentials are single-source. No amount of engineering conjures them, and the platform is designed to refuse rather than approximate. A delay here delays a first filing regardless of code readiness.

**Breadth as a liability**

The lifecycle span is the differentiator and also the risk. Fifteen stages at uneven depth invites a demo that impresses and a pilot that finds the thin places. The audit's judgment — make *one* journey complete before broadening — is the right sequencing, and it means near-term progress will look narrower than the eventual scope.

**Incumbent response**

The largest RIM vendors began shipping AI agents in 2026. Their advantage is distribution into accounts this platform must win from outside. The window in which their AI is a workflow bolt-on rather than a drafting engine is finite and not controlled by this company.

**Validation is a recurring cost**

Regulated customers require vendor validation evidence, refreshed as the product changes. This is an ongoing operational burden on every release, not a one-time engineering task, and it structurally slows shipping velocity after GA.

**No commercial proof yet**

This paper contains no revenue, pipeline, or customer figures because none are derivable from the codebase. The technical case stands on its own; the commercial case is a separate diligence item and should be requested separately.

### What an investment is buying

A lifecycle platform in a market of point tools, built against a standards migration that has forced every sponsor's stack open, with the domain depth already paid for and the remaining work scoped by an independent audit rather than estimated internally. Specifically: 215 service domains and 931 governed tables spanning grant funding to post-approval change; deterministic engines whose outputs are citable rather than generated; provenance enforced structurally rather than by policy; and a demonstrated rate of closing a named defect list.

What it is not buying is a finished product. The platform is pre-GA, its regulated core is mid-hardening, several lifecycle stages are thin, and the compounding data moat is designed but not collecting.

IN SUMMARY

What has been built is the hard part: a governed chain from source evidence to transmitted submission and beyond it in both directions, with provenance enforced structurally rather than by policy, across both regulatory worlds. What was wrong with it has been independently found, published here rather than buried, and is being closed at a measurable rate. The market is fragmented, the migration has forced it open, and the architecture that spans it exists in code today — the question is execution against a known list, on a clock, and that is the risk being priced.

Prepared August 2026 against the `concept2cure-v2` engineering branch. Quantitative figures derived directly from the repository at that revision; audit figures from an independent 15-domain readiness assessment dated 10 August 2026, with remediation claims in Section 24 re-verified in the code for this document. A capability inventory with regeneration commands is maintained alongside this paper. 